Issue	**1**
Factory ID	**25650**
Date of Issue	**16-Jun-2026**

Amtex Ltd.

Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

5-Pullian Ismail Road, Tehsil Faisalabad City,
Faisalabad
Punjab Pakistan

Latitude: [31.45965], longitude: [73.0738]
Date of inspection: 13-Apr-2026

Electrical Safety Inspection Report

Observations & Actions



Authors	**Azka Riaz**
Reviewed by	**Majid Khan**
Approved by	**Colm Quinn**

Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 13-04-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

Contents

1. Executive Summary 5
2. Building Extents 7
   - 2.1 Building Use 9
   - 2.2 Utility Source 10
   - 2.3 Backup source 11
3. Documentation 12
4. Electrical Safety Measures 13
5. Remediation Support 14
   - 5.1 Engineering Support 14
6. Diagrams & Drawings 15
   - 6.1 Sub-station room layout drg 15
   - 6.2 Single Line Diagram (SLD) 16
   - 6.3 Electrical layout drawing 17
7. DBs & Elec Protection Systems 18
   - 7.1 Circuit breakers 18
   - 7.2 Hot Spots 21
   - 7.3 Earthing system 25
   - 7.4 Panel/cabling/busbar 29
   - 7.5 Loop Connection 34
   - 7.6 Multiple Connections 36
   - 7.7 Openings 38
   - 7.8 Adequate accessibility 40

7.9 Lint & dust inside panel board....42
7.10 Lightning Protection System (LPS)....44
8. Fire Alarm & Detection System (FADS)....46
8.1 Non-functional or incomplete....46
9. Emergency Lighting....49
9.1 Provided but non-functional....49
9.2 Provided but insufficient....51
9.3 Independent power source....54
9.4 General Lighting....55
10. Generator, Transformer & Sub-Station Room....57
10.1 Room separation....57
10.2 Battery....59
10.3 Back-up power supply....62
11. Cabling & Wiring....64
11.1 Cable support and protection....64
11.2 Cable trench and channel cleanliness....67
11.3 Splicing of cables....69
11.4 Colour coding....72
11.5 Fixtures....74
12. Assessment Limitations and Assumptions....77

1. Executive Summary

On 13-Apr-2026, Azka Riaz, Muhammad Bilal Butt and Hamza Talib on behalf of the Pakistan Accord carried out an initial electrical inspection for Amtex Ltd., a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Naveed Aslam (Compliance Manager), Mr. Faisal Ameer (Admin), Mr. Tajammul Hussain (Electrical Engineer) of Amtex Ltd. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were present.

The factory operates in general shift for all the activities with approximately 500 workers operating in the factory, with 20 workers being female.

The factory comprises two main buildings and three ancillary structures owned by Amtex Ltd. During the inspection, the buildings highlighted in Figure 1.1 were inspected, with the site boundary indicated in red.

Building 01 is 2 storey (Ground + 01). The building is used for solar on the roof, stitching, packing, checking offices on the 1st floor, offices, cutting, folding, fabric store, processing, sample room, wadding, and quilting on the ground floor.
Building 02 is a single storey (Ground only) structure. The building is used for solar on the roof, printing, cutting, finishing, packing, warehouse, store, mosque on the ground floor.
The other buildings comprises three structures.

- Ancillary toilet block is a single storey (Ground only). It is used for washrooms.
- Ancillary dormitory 01 is a single storey (Ground only). It is used for restrooms.
- Ancillary dormitory 02 is a single storey (Ground only). It is used for restrooms.

A summary of the main concerns in relation to life safety systems identified during the inspection (non-exhaustive) that should be actioned is as follows:

• **Abnormal Temperature:** Within electrical panels, abnormal temperatures were recorded at various locations (refer to section 7.2 for further details). Hotspots within panels can cause short circuits and fire due to insulation damage and overheating of cables or electrical components.
• **Fire Alarm and Detection System (FADS):** The facility lacks a fire alarm control panel, relying on inadequate standalone detectors and 220V hooters with manual push buttons, which were found broken and non-functional, posing a significant life safety risk in the event of a fire emergency.
• **Emergency lighting System:** The emergency lighting system exhibited several deficiencies, including inadequate coverage along egress routes, emergency staircases, and assembly areas, as well as non-functional luminaires, hindering safe evacuation during power outages and increasing the risk of accidents.
• **Earthing system:** The earthing system was found to be inadequate in several areas of the facility. The main LT panel and most distribution boards lacked earthing conductors, and many electrical machines were not provided with proper earth connections. The lack of a proper earthing system

poses significant safety risks, such as electric shocks, fire hazards, and potential equipment damage.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Satellite view of the factory premises; Red dashed line indicates site boundary and highlighted blue area indicates buildings inspected

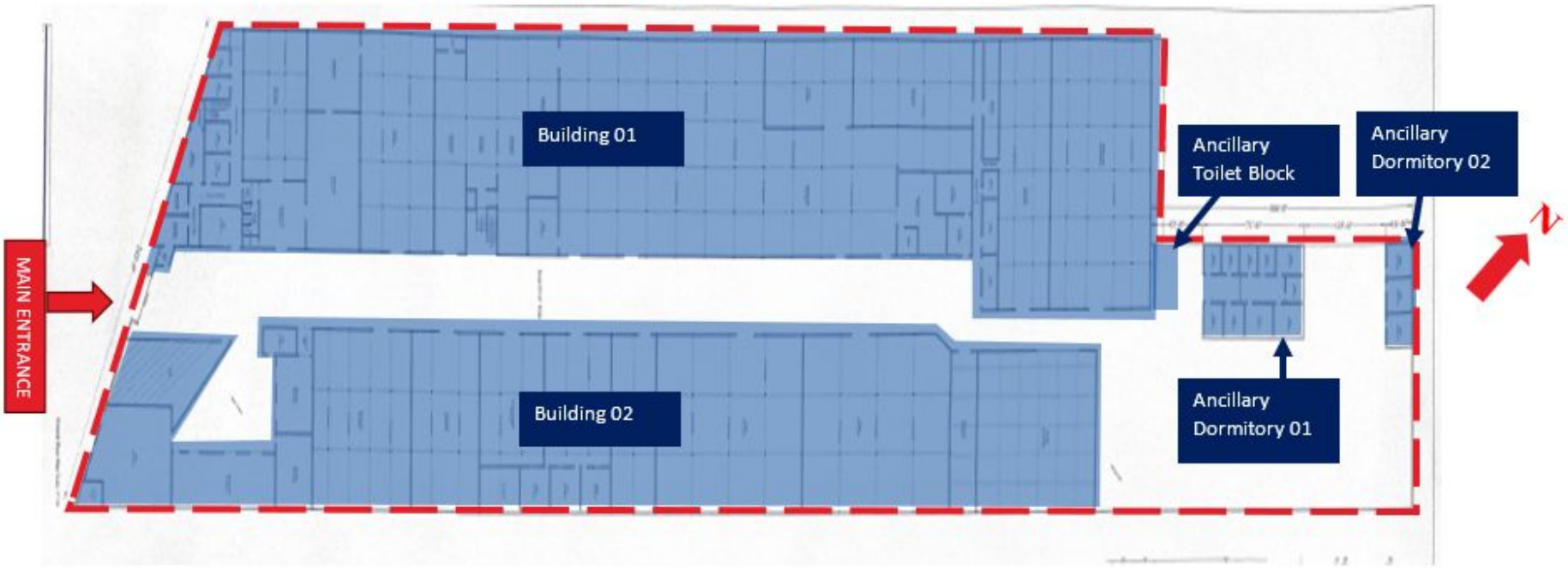


Figure 1.2. Site Layout Plan of the Factory Premises - Red dashed line indicates site boundary and highlighted areas indicates buildings inspected

2.1 Building Use

The factory comprises two main buildings and three ancillary structures. The floor-wise activity details are as follows:

Building 01:
L00 – Ground Floor: Offices, cutting, folding, fabric store, processing, sample room, wadding, and quilting.
L01 – First Floor: Stitching, packing, checking, and offices.
L99 – Roof: Solar panels.

Building 02:
L00 – Ground Floor: Printing, cutting, finishing, packing, warehouse, store, and mosque.
L99 – Roof: Solar panels.

Ancillary Buildings:

Ancillary Toilet Block:
L00 – Ground Floor: Washrooms.

Ancillary Dormitory 01:
L00 – Ground Floor: Restrooms.

Ancillary Dormitory 02:
L00 – Ground Floor: Restrooms.

2.2 Utility Source

The facility receives a single 11 kV utility supply connection with a sanctioned load of 503 kW from the Faisalabad Electric Supply Company (FESCO) as its primary power source, in the utility area of Building 1. This area houses one HT panel and one oil-type step-down transformer rated 800 kVA owned and maintained by the facility. The transformed power is fed to the main Low Tension (LT) panels located in the same area, from where it is distributed throughout the facility via distribution boards. Additionally, a 242 kW solar power system is also synchronized with the primary power source.

2.3 Backup source

The facility has one low-voltage 800 kVA diesel generator serving as a backup power source and one 100 kVA low-voltage diesel generator as a standby source. The fire alarm and detection system, along with the emergency lighting system, were equipped with built-in battery backup supplies. Additionally, the general lighting of office areas was supplied with a UPS backup supply.

3. Documentation

This table shows the details of the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No	-	-
2	Single Line Diagrams	No	-	-
3	Fire Alarm Drawing	No	-	-
4	Earth Test Report	No	-	-
5	Thermographic Test Report	No	-	-
6	Insulation Resistance Test Report	No	-	-
7	LPS Risk Assessments	No	-	-
8	Periodic Maintenance Record	No	-	-
9	Transformer Oil Testing Report	No	-	-

4. Electrical Safety Measures

During our electrical site inspection of Amtex Ltd., a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Amtex Ltd.. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician. Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202604-065704

Timeline	Remediation Plan
Medium Term CAP-0078095	Factories should begin remediation of CAP items, such as cleaning, cable organization, and basic corrections, using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202604-065609

Timeline	Remediation Plan
Medium Term CAP-0077786	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077787	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202604-065599

Timeline	Remediation Plan
Medium Term CAP-0077784	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077785	Implement a regular review and update process for the single-line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202604-065590

Timeline	Remediation Plan
Medium Term CAP-0077789	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077790	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey, the circuit breakers installed throughout the facility, inside the electrical panels and distribution boards, were inspected.

i. It was observed that the circuit breakers were installed improperly, without proper fixing supports and protective enclosures. This condition poses a risk of electric shock, short circuit, and potential fire hazards due to exposure to live parts and environmental conditions. This issue was observed specifically in the substation room, solar inverter room, garments cutting area, coating department, and wrapping machine distribution board.

ii. It was found that a single-phase load was fed from a three-pole breaker. Using a single phase of a three-pole breaker can lead to unbalanced load conditions. This can cause the breaker to trip unexpectedly or, in some cases, fail to trip when necessary, compromising the safety of the system. This issue was observed specifically in the LT panel, quilting & wadding distribution board, coating machine distribution board, and curing machine electrical connections.

iii. It was found that some loads inside the distribution boards were directly connected without overcurrent protection or a circuit breaker. This issue was specifically observed for the emergency lighting circuit connection from the stitching unit-3 distribution board.

iv. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes.
a. In the LT panel, 200A circuit breakers were connected to 25 mm^2 and 16 mm^2 PVC cables.
b. In the stitching unit-3 distribution board, a 100A circuit breaker was connected to 4 mm^2 PVC cables.
c. In the coating machine distribution board, a 300A circuit breaker was connected to 35 mm^2 PVC cables.
d. In the made-up stitching hall distribution board, a 400A circuit breaker was connected to 25 mm^2 PVC cables.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202604-065602

Timeline	Remediation Plan
Medium Term CAP-0077995	i. Secure all loosely installed circuit breakers with proper mounting and enclose them in compliant metallic or insulated enclosures, in accordance with BS 7671 Section 416 (Barriers and Enclosures for Basic Protection).
Medium Term CAP-0077996	ii. Remove single-phase connections from three-pole breakers and replace them with appropriate single-pole breakers, ensuring compliance with standard current-carrying capacities and preventing load imbalance.
Medium Term CAP-0077997	iii. Install appropriate protective devices, such as circuit breakers or fuses, for all loads to ensure safe operation and compliance with electrical safety regulations.
Medium Term CAP-0077998	iv. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - 200 A circuit breaker connected to 25 mm² and 16 mm² PVC cables, LT panel



Figure 2.2 - Circuit breakers were mounted improperly without



Figure 2.3 - Emergency lighting load connected without overcurrent protection, stitching unit-3 distribution board



Figure 2.5 - Circuit breaker installed improperly in the coating department

protective enclosure in substation room



Figure 2.4 - 400A circuit breaker was connected to 25 mm² PVC cables, made-up stitching distribution board



Figure 2.6 - Single phase fed from three-pole breaker found in the LT panel

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:
i. 129°C – found at cable joint in Quilting area distribution board
ii. 126°C – found at incoming circuit breaker cable connection in Solar inverter room (general lighting)
iii. 113°C – found at magnetic contactor cable connection in Steamer distribution board (outside Garment 2 finishing area)
iv. 105°C – found at circuit breaker cable connection in Quilting area distribution board
v. 103°C – found at magnetic contactor cable connection in RO plant distribution board
vi. 94.5°C – found at circuit breaker cable connection in Made-up cutting area distribution board
vii. 91.9°C – found at circuit breaker in Solar inverter room distribution board-3
viii. 86.8°C – found at magnetic contactor cable connection in Steamer distribution board (outside Garment 2 finishing area)
ix. 83.9°C – found at circuit breaker cable connection in Quilting area distribution board
x. 82.1°C – found at circuit breaker terminal in Made-up cutting area distribution board
xi. 80.2°C – found at circuit breaker cable connection in Garments 2 finishing area distribution board
xii. 79.1°C – found at cable joint in cable tray (Stitching Hall 2 near HTL machine)

xiii. 72.4°C – found at magnetic contactor cable connection in Steamer distribution board (outside Garment 2 finishing area)
xiv. 71.8°C – found at circuit breaker cable connection in Stitching Hall 2 distribution board
xv. 68.4°C – found at circuit breaker cable connection at main panel (main breaker, Wadding plant

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC-insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment. F-202604-065597

Timeline	Remediation Plan
Immediate CAP-0077999	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0078000	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0078001	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit the thermographic test report to the Pakistan Accord for review.

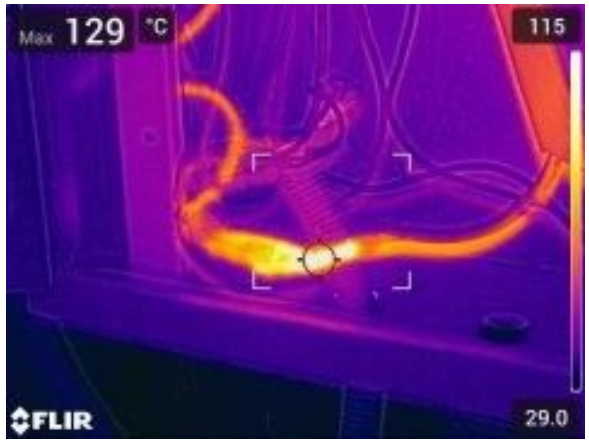


Figure 2.7 - Hot spot found on the cable joint in Quilting area distribution board



Figure 2.8 - Hot spot found on the cable joint in Quilting area distribution board



Figure 2.9 - Hot spot found on the incoming circuit breaker cable connection in solar inverter room



Figure 2.10 - Hot spot found on the incoming circuit breaker cable connection in solar inverter room



Figure 2.11 - Hot spot found on the magnetic contactor cable connection in Steamer distribution board



Figure 2.12 - Hot spot found on the magnetic contactor cable connection in Steamer distribution board



Figure 2.13 - Hot spot found on the circuit breaker cable connection in Quilting area distribution board



Figure 2.14 - Hot spot found on the circuit breaker cable connection in Quilting area distribution board

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. No earthing system design documentation was provided by the factory, indicating a lack of verification of the earthing system’s compliance and ongoing performance in accordance with relevant standards.

ii. The earth pits were inaccessible and unsuitably located in the generator's area.

iii. The earthing cables were not appropriately sized according to the relevant earthing standards. This issue was observed specifically in the generator and main LT panel.

iv. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables, and many electrical machines did not have proper earth connections. This issue was observed specifically in the stitching unit 3 distribution board, the made-up cutting area distribution board, and the grey godown distribution board, as well as in stitching machines in the garments stitching hall in building 2, the stitching hall on the first floor of building 1, and the cutting machine in the sample room on the ground floor of building 1.

v. Equipotential bonding for the cable trays and metallic panel covers was missing. This issue was observed specifically in the quilting & wadding distribution board and the made-up stitching hall distribution board.

vi. The earth pit testing and maintenance records were not provided.

The lack of a proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202604-065608

Timeline	Remediation Plan
Medium Term CAP-0077987	i. Design and evaluate the facility's earthing system, and develop layout drawings in accordance with BS 7671 18th Edition + A2 (2022). Submit the earthing and neutral layout drawings to the Pakistan Accord for review.

Medium Term CAP-0077988	ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.
Medium Term CAP-0077989	iii. Replace all earthing cables with appropriately sized cables as per the requirements of BS 7671 18th Edition + A2 (2022) Appendix 4, Section 5. Verify that all earthing conductors meet the minimum cross-sectional area specified.
Medium Term CAP-0077990	iv. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0077991	v. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Medium Term CAP-0077992	vi. Implement periodic earth testing per BS 7671 Regulation 643.7.1 to verify grounding effectiveness and submit testing record to Pakistan Accord for review.
Longer Term CAP-0077993	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.15 - Earthing cable missing in the stitching unit-3 distribution board



Figure 2.16 - Earthing cable missing in the made-up cutting distribution board



Figure 2.17 - Earthing connection missing for the stitching machines on the first floor, building 1



Figure 2.18 - Equipotential bonding missing for panel doors in the made-up stitching distribution board



Figure 2.19 - Earthing pit inaccessible and unsuitably located in the generator area



Figure 2.20 - Undersized earthing cables observed in the main LT panel

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables, and busbars throughout the facility.

i. It was observed in the substation rooms that electrical components were mounted on a wooden support without protective enclosures, posing a fire risk due to the combustible nature of the material and potential overheating.

ii. It was observed that electrical components were not suitable for the environmental conditions in which they were installed. This issue was observed specifically in the water pump distribution boards and gas steamer electrical panels installed outside the buildings.

iii. Several electrical panels exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements. This issue was observed specifically in the RO plant distribution board, the lamination machine panel, and on the neutral bar in the boundary lights electrical enclosure.

iv. There were visible signs of damage to panels, cables, and busbars, including broken electrical enclosures, broken panel glass doors, faulty phase indicators, and signs of burning. This issue was observed specifically in the coating department distribution board, printing area distribution board, cotton passion distribution board, garments stitching distribution board, and garments cutting electrical enclosure.

v. Phase separators were missing in LT panels and most of the distribution boards, posing the risk of accidental contact between phases.

vi. Inadequate cable dressing was observed, with conductors not properly arranged, secured, or terminated neatly. This resulted in untidy wiring, poor segregation between circuits, and unnecessary strain on terminations, which can lead to loose connections, overheating, and difficulty during inspection or fault-finding. This issue was observed specifically in the compressor area and the reception area distribution board.

The utilization of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202604-065594

Timeline	Remediation Plan

Immediate CAP-0078013	i. Install the electrical components on non-combustible supports and enclose them in compliant metallic or insulated enclosures, in accordance with BS 7671 Section 416 (Barriers and Enclosures for Basic Protection).
Medium Term CAP-0078014	ii. Identify and replace electrical panels, cables, and busbars that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0078015	iii. Replace or refurbish corroded electrical panels and components to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0078016	iv. Inspect and repair or replace all damaged panels, cables, and busbars to ensure the integrity of the electrical system in accordance with Section 522 of BS7671 18th Edition + A2 (2022). Replace the defective indication lights with new and reliable components.
Medium Term CAP-0078017	v. Phase separators should be installed between adjacent live conductors of different phases within distribution boards to ensure safe segregation and protection against direct contact, in accordance with BS 7671 Regulation 416.2.
Medium Term CAP-0078018	vi. Ensure proper cable management by organizing and securing cables within the panels using cable ties, clips, or ducting. Verify that all components and connections are securely mounted, neatly arranged, and labelled to improve safety, ease of maintenance, and compliance with electrical standards such as BS 7671.
Longer Term CAP-0078019	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.21 - Improper cable dressing observed in the reception area distribution board



Figure 2.22 - Electrical components mounted on a wooden support without protective enclosure, substation room



Figure 2.23 - Faulty phase indication lights observed in the printing area distribution board



Figure 2.24 - Improper cable dressing observed in the compressor area



Figure 2.25 - Sign of burning observed on the circuit breaker cable connection in the garments stitching distribution board



Figure 2.27 - Corrosion observed on the neutral bar in the boundary lights electrical enclosure



Figure 2.26 - Non-IP rated water pump distribution board installed outside building 1



Figure 2.28 - Phase separators missing in the coating department distribution board



Figure 2.29 - Corrosion observed on the RO plant distribution board



Figure 2.30 - Garments cutting electrical enclosure found broken

7.5 Loop Connection

This relates to the entire facility.

During the inspection, loop connections were identified within the electrical panels. This issue was observed specifically in the distribution boards located in the garments finishing area, reception area, and admin office area, as well as circuit breakers installed in the solar inverter room and quilting and wadding area.

In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202604-065601

Timeline	Remediation Plan
Medium Term CAP-0077932	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.31 - Loop connections found in the admin distribution board



Figure 2.32 - Loop connections found in the garments finishing distribution board



Figure 2.33 - Loop connections found in the reception area distribution board



Figure 2.34 - Loop connections found in circuit breakers installed in the solar inverter room

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility. This issue was observed specifically in the main LT panels, the solar inverter distribution board #3, the made-up cutting area distribution board, the quilting and wadding distribution board, and the reception distribution board.

In these cases, multiple circuits were being protected by a single circuit breaker.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202604-065592

Timeline	Remediation Plan
Medium Term CAP-0078043	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).



Figure 2.35 - Multiple connections found in the made-up cutting area distribution board



Figure 2.36 - Multiple connections found in the quilting & wadding distribution board



Figure 2.37 - Multiple connections found in the reception distribution board



Figure 2.38 - Multiple connections found in the main LT panel

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings and missing baseplates were observed in the panels and distribution boards. This issue was observed specifically in the main LT panels, solar inverter distribution board #3, stitching unit-3 distribution board, made-up stitching hall distribution board, garments finishing area distribution board, quilting and wadding distribution board, and coating department distribution board.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202604-065596

Timeline	Remediation Plan
Medium Term CAP-0078003	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0078004	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.39 - Unsealed openings observed in the solar inverter distribution board #3



Figure 2.40 - Unsealed openings observed in the stitching unit-3 distribution board



Figure 2.41 - Baseplate missing in the made-up stitching hall distribution board



Figure 2.42 - Unsealed openings observed in the coating department distribution board

7.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At numerous locations, it was found that the electrical panels do not possess the required space for inspection or easy access to the panels and their components. This issue was observed specifically in the printing area, grey godown, and cargo lift distribution board on the rooftop of building 1.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202604-065593

Timeline	Remediation Plan
Medium Term CAP-0078033	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.43 - Inadequate accessibility for the printing area distribution board



Figure 2.44 - Inadequate accessibility for the cargo lift distribution board on the rooftop, building 1



Figure 2.45 - Inadequate accessibility observed for the grey godown distribution board

7.9 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards, and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels, and trunking systems. This issue was observed in the generator control panel, grey godown distribution board, coating area distribution board, lamination machine electrical panel, fabric store, and stitching electrical enclosures, as well as in cable trays within the stitching halls on both floors of building 1.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202604-065606

Timeline	Remediation Plan
Medium Term CAP-0077904	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0077905	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.46 - Lint and dust observed in the stitching hall on the ground floor of building 1



Figure 2.47 - Lint and dust observed in the stitching hall on the first floor of building 1



Figure 2.48 - Lint and dust found in the coating area distribution board



Figure 2.49 - Lint and dust found in the generator control panel

7.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

It was observed that a lightning protection system was provided; however, there was no study available to confirm its adequacy, and no risk assessment was available.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202604-065591

Timeline	Remediation Plan
Medium Term CAP-0078041	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment according to BS EN 62305 (Protection against Lightning) standard. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0078042	If the risk assessment concludes that an LPS is necessary, then hire a qualified engineer to perform an adequacy study of the existing LPS. This study should evaluate whether the current system meets the necessary design and performance standards according to BS EN 62305. If the study finds that the current LPS is inadequate, make the necessary upgrades to bring the system up to standard.



Figure 2.50 - Lightning arrestor installed on the rooftop of building 1

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

The facility lacks a fire alarm control panel, relying on inadequate standalone detectors and 220V hooters with manual push buttons, which were found broken and non-functional, posing a significant life safety risk in the event of a fire emergency.

The use of standalone detectors and alarms, along with the absence of an appropriate control panel, constitutes non-compliance with NFPA 72 standards, compromising the system's effectiveness and reliability.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202604-065604

Timeline	Remediation Plan
Immediate CAP-0077807	Develop a comprehensive emergency response plan for fire incidents in the factory. This plan should include implementing an alert mechanism, such as a Public Address (PA) system or a manual alarm system, to promptly notify all personnel in the facility. The plan must allow for an orderly evacuation process to ensure the safety of all workers in the event of a fire.
Medium Term CAP-0077808	i. Submit design documentation for the new replacement of FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0077809	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.

Longer Term CAP-0077810	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0077811	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0077812	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0077813	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.

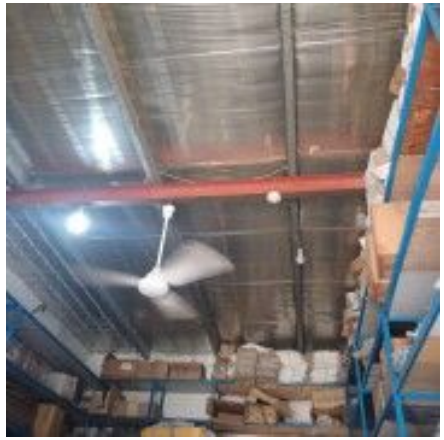


Figure 2.51 - Standalone smoke alarm installed in the accessory store



Figure 2.52 - Fire alarm and detection coverage missing in the electrical room



Figure 2.53 - Alarm hooter found broken and non-functional in garments 2 cutting entrance area



Figure 2.54 - Fire detection coverage missing in cooking area near admin offices



Figure 2.55 - Alarm hooters not provided in stitching store area

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection, it was observed that emergency lighting was provided but not functional in the warehouse and the garments cutting areas in both buildings 1 and 2.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202604-065598

Timeline	Remediation Plan
Medium Term CAP-0077882	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0077883	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.56 - Non-functional emergency light found in the garments cutting area, building 1



Figure 2.58 - Non-functional emergency light found in the garments cutting area, building 2



Figure 2.57 - Non-functional emergency light found in the warehouse

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection, the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency lighting and exit directional signs coverage. This issue was observed in the generator area, solar inverter room, quilting and wadding area, coating department, garments stitching passage, cotton passion store & packing area, and internal stairs of building 1. Additionally, the layouts for emergency lighting were missing. The absence of emergency lighting layouts throughout the facility raises concerns regarding the adequacy of illumination provided during power failures or emergencies.

ii. The backup power supply to the emergency luminaires was insufficient.

iii. A central UPS was utilized to provide backup power to the normal lighting in the admin offices of building 1; however, battery autonomy calculations were not provided, non-fire-rated cables were used for the emergency lighting circuits, and the UPS placement does not comply with the standard requirements.

iv. External emergency lighting was not provided, particularly in the external emergency stairs and the assembly area, preventing people from reaching a place of safety outside the facility during nighttime evacuations.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202604-065603

Timeline	Remediation Plan
Medium Term CAP-0077830	i. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Prepare and submit the emergency light design and layouts to the Pakistan Accord for review. Install additional emergency luminaires and exit directional signs in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.

Medium Term CAP-0077831	ii. Upgrade the backup power supply to ensure it can adequately support all emergency luminaires during a power failure. This includes installing additional battery packs or generators as necessary to comply with the Building Code of Pakistan Fire Safety Provisions 2016.
Medium Term CAP-0077832	iii. Prepare and submit verified UPS battery autonomy calculations to demonstrate compliance with emergency lighting duration requirements, replace non-fire-rated cables with compliant fire-rated cabling for emergency lighting circuits, and relocate or enclose the UPS in accordance with applicable fire and electrical safety standards.
Medium Term CAP-0077833	iv. Install external emergency luminaires to enable safe evacuation during hours of darkness, all the way to a safe place outside the building(s).
Longer Term CAP-0077834	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.





Figure 2.59 - External emergency lighting not provided in the stitching store area



Figure 2.60 - Emergency lights and illuminated exit directional signs not provided in internal stairs, building 1



Figure 2.61 - Improper placement of UPS in admin offices area, building 1



Figure 2.62 - Emergency lighting and exit directional signs not provided in garments stitching passage



Figure 2.63 - Emergency light missing in the generators area

Figure 2.64 - Emergency lighting and exit directional signs not installed in the cutting department

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs should be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202604-065605

Timeline	Remediation Plan
Medium Term CAP-0077795	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately upon failure of general lighting power as required by the Building Code of Pakistan Fire Safety Provisions 2016. Emergency lighting shall be powered from a source independent of that supplying the normal lighting.
Longer Term CAP-0077796	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

9.4 General Lighting

The section refers to the entire facility.

During the inspection, the general lighting throughout the facility was assessed.

It was observed that the general lighting was controlled directly from the circuit breakers within the Distribution Boards rather than through standard light switches. This issue was specifically observed in the stitching hall on the first floor of building 1.

Insufficient general lighting in several locations can increase the risk of accidents and impede safe evacuation during emergencies, while controlling general lighting via breakers instead of light switches poses operational inefficiencies and safety hazards. This setup can delay emergency responses and heighten the risk of electrical faults or fires due to frequent breaker use. Addressing these issues is crucial for enhancing everyday safety and ensuring effective emergency preparedness in the facility.F-202604-065610

Timeline	Remediation Plan
Medium Term CAP-0077793	Reconfigure the electrical control system to ensure that general lighting is operated via standard wall-mounted light switches instead of circuit breakers. This involves installing appropriate switches and rewiring the control circuits as per the National Electrical Code (NEC) and other relevant electrical safety standards. Ensure that all switches are easily accessible and clearly labelled to enhance operational efficiency and safety.



Figure 2.65 - General lighting was controlled directly from the circuit breakers in the stitching hall on the first floor, building 1

10. Generator, Transformer & Sub-Station Room

10.1 Room separation

During the inspection of the generator, transformer, and substation rooms, the physical separation between these critical areas was assessed. The barriers, walls, and partitions intended to isolate each component were reviewed for adequacy and compliance with safety standards.

It was observed that adequate and code-compliant separation was not maintained between oil-type transformers, the HT panel, and the LT panels. In several locations, barriers were either absent or insufficient, with visible gaps and openings that could allow the spread of fire or electrical faults between adjacent sections. Additionally, some partitions were constructed from materials that do not meet the required fire-resistance ratings.

This condition was noted in the substation room in building 1.

The lack of proper segregation significantly increases the risk of fault escalation, fire propagation, equipment damage, and potential injury to personnel. Oil-type transformers require proper fire-rated separation, which was found to be inadequate at the time of inspection. F-202604-065595

Timeline	Remediation Plan
Medium Term CAP-0077985	Provide immediate fire risk mitigation by installing temporary fire-resistant barriers to segregate oil-type transformers from HT/LT panels and other equipment. Seal all gaps and openings, and ensure safe clearances are maintained to minimize the risk of fire and fault propagation.
Longer Term CAP-0077986	Redesign and upgrade the substation layout to ensure full compliance with applicable standards (e.g., BS 7671, NFPA guidelines), including dedicated, fire-rated compartments for oil-type transformers, appropriate segregation between HT and LT panels, and use of compliant fire-rated construction materials with no unprotected openings. Implementation of a periodic inspection and maintenance program to ensure long-term integrity of all segregation measures.



Figure 2.66 - Oil-type transformer, HT panel and LT panels housed in the same room in building 1, violating separation requirements

10.2 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

The following observations were noted with the battery system:

i. Noticeable spills of battery fluid were observed, which can lead to corrosion and pose a chemical hazard.

ii. Several battery terminals were exposed and not properly insulated, increasing the risk of electrical shock or short circuits.

iii. The placement of some batteries did not comply with safety regulations, potentially hindering safe access and proper ventilation.

These issues were observed with the generator batteries and UPS batteries placed in the RO plant room and the admin offices area.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202604-065600

Timeline	Remediation Plan
Medium Term CAP-0077862	i. Clean any battery fluid spills using appropriate neutralizing agents and safety equipment to prevent corrosion and chemical hazards. Inspect and repair any affected battery cells or enclosures to prevent further leakage.
Medium Term CAP-0077863	ii. Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Medium Term CAP-0077864	iii. Reposition batteries that are improperly placed to ensure they comply with safety regulations, allowing for safe access and adequate ventilation. Mark and clear pathways around battery installations to ensure easy and safe access for maintenance and emergencies.
Longer Term CAP-0077865	i. Implement a regular maintenance schedule to monitor and address potential leaks, ensuring that any spills are promptly managed.

Longer Term CAP-0077866	ii. Establish a routine inspection and maintenance protocol to check and maintain proper insulation of all battery terminals regularly.
Longer Term CAP-0077867	iii. Conduct regular audits of battery placements to ensure ongoing compliance with safety standards and regulations.



Figure 2.67 - Improperly placed UPS batteries in the RO plant room



Figure 2.69 - Exposed terminals and spillage observed on the UPS batteries, RO plant



Figure 2.68 - Exposed terminals and spillage observed on the generator batteries

10.3 Back-up power supply

During the assessment of the backup power supply system, an inspection was conducted on the generators, battery backups, and auxiliary power sources.

Both backup and standby generators were observed with signs of inadequate maintenance, including corrosion, deteriorated components, and oil leakages.

Maintaining a reliable backup power supply system is crucial for ensuring continuous operations and safety during power outages or emergencies. Inadequate maintenance of generators can lead to failures during critical moments, jeopardizing essential services and potentially causing disruptions or safety hazards.F-202604-065607

Timeline	Remediation Plan
Medium Term CAP-0077847	Conduct comprehensive maintenance, repair or replace deteriorated components, and rectify all oil leakages in accordance with manufacturer recommendations.
Longer Term CAP-0077848	Establish a regular maintenance schedule to ensure all generators are routinely serviced, including timely replacement of components and fuel filters.



Figure 2.70 - Corrosion and radiator leakages observed in the backup supply generator



Figure 2.71 - Corrosion and oil leakages observed in the standby supply generator



Figure 2.72 - Corrosion and deteriorated components observed in the backup supply generator

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

The following observations were noted:

i. Proper supports were not provided for cables; they were laid directly on the ground and hung along the walls. Additionally, several locations had open and exposed electrical connections. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure. This issue was observed specifically in the substation room, fabric store, and on the rooftop of building 1.

ii. Loose connections identified in the grey godown can generate sparks and potentially cause a fire. It is essential to avoid loose connections to prevent electrical hazards.

iii. It was observed that cables were installed without proper cable containment (such as cable trays, conduits, or ducting) in the storage areas. Inadequate containment and support may lead to mechanical damage, sagging, and stress on the cables, potentially resulting in insulation failure or faults. This issue was observed specifically in the coating department.

iv. Cable trays and conduits were found damaged and inadequately supported at multiple locations, including on the rooftop of building 16, in the stitching unit-3, coating department, grey down, cotton passion stitching area. Such deterioration and insufficient support can lead to cable damage, overheating, and potential electrical faults or fire hazards.F-202604-065586

Timeline	Remediation Plan
Medium Term CAP-0078073	i. Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).

Medium Term CAP-0078074	ii. Tighten and secure all loose connections immediately to eliminate the risk of sparks and potential fires. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 526.1 for ensuring secure connections).
Medium Term CAP-0078075	iii. Provide proper conduit or trunking for all exposed cables to ensure mechanical protection and compliance with electrical safety standards.
Medium Term CAP-0078076	iv. Repair or replace the damaged cable trays and ensure proper secure mounting according to BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support.
Longer Term CAP-0078077	Establish a preventive maintenance program to regularly inspect and maintain cable installations, including proper support, secure terminations, and provision of suitable containment systems. Train personnel on safe wiring practices and ensure compliance with relevant electrical standards to sustain system integrity and reduce potential hazards.



Figure 2.73 - Damaged cable tray observed in the stitching unit-3



Figure 2.74 - Loose connections observed in the grey godown



Figure 2.75 - Inadequately supported cable tray observed in the cotton passion stitching area



Figure 2.76 - Inadequately supported cables observed with the transformer



Figure 2.77 - Open wiring was observed in the storage areas without cable conduits, coating department

11.2 Cable trench and channel cleanliness

This is related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of fluff and dust, which poses a serious fire risk. This issue was observed specifically in the substation room cable trench.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimize fire risks associated with accumulated debris.F-202604-065587

Timeline	Remediation Plan
Medium Term CAP-0078062	Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Longer Term CAP-0078063	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.78 - Cleanliness of cable trench not maintained in the substation room



Figure 2.79 - Cleanliness of cable trench not maintained in the substation room

11.3 Splicing of cables

This relates to the whole facility.

During the inspection, the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed specifically in the LT panels, solar inverter distribution board #2, RO plant area distribution board, quilting & wadding distribution board, garments finishing area distribution board, reception distribution board, and garments stitching area distribution board.

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was observed specifically in the LT panels, quilting & wadding distribution board, stitching unit-3 cable tray, wrapping area distribution board, made-up cutting area distribution board, made-ups stitching hall distribution board, garments finishing area distribution board, and solar inverter room.

iii. Non-compliant and low-quality PVC connectors used in the stitching hall distribution board could be a hazard of a potential short circuit due to their poor insulation.

For any future electrical work, the termination of stranded cables must be done with cable lugs, and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202604-065585

Timeline	Remediation Plan
Medium Term CAP-0078086	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).

Medium Term CAP-0078087	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Medium Term CAP-0078088	iii. Replace the low-quality PVC connectors with standard quality terminal blocks in accordance with BS7671 18th Edition + A2 (2022).
Longer Term CAP-0078089	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Longer Term CAP-0078090	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0078091	iii. Establish a plan to replace all low-quality PVC connectors with compliant electrical terminal blocks and integrate this into regular maintenance to ensure ongoing safety and compliance.



Figure 2.80 - Cable terminations without lugs and spliced cables found in the solar inverter room



Figure 2.81 - Cable terminations without lugs and spliced cables found in the reception distribution board



Figure 2.82 - PVC tape joints found in the stitching unit-3 cable tray



Figure 2.83 - Improperly spliced cable joints with PVC tapes observed in the wrapping area distribution board



Figure 2.84 - Low-quality PVC connector found in the stitching hall distribution board

11.4 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

This issue was observed in LT panels and distribution boards located in the stitching, cutting, coating, garments finishing, and reception areas.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202604-065589

Timeline	Remediation Plan
Medium Term CAP-0077917	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0077918	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.85 - Incorrect color coding observed on the neutral bar in the LT panel



Figure 2.86 - Incorrect color coding observed in the stitching distribution board



Figure 2.87 - Incorrect color coding observed in the coating area distribution board



Figure 2.88 - Incorrect color coding observed in the garments finishing area distribution board

11.5 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles found installed outside the building. Receptacles placed in unsuitable locations, such as non-IP-rated sockets installed outdoors or exposed to the elements, are at higher risk of moisture ingress and corrosion. This can lead to electrical short circuits or fires, posing a severe safety hazard.

ii. Power receptacles mounted on combustible wooden ply: Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety. This issue was observed in the substation room, solar inverter room, quilting & wadding area, coating department, and garments stitching hall.

iii. Broken and exposed power receptacles: Damaged receptacles with exposed parts can lead to loose connections and increased electrical resistance. This condition can cause overheating, potentially igniting surrounding materials and increasing the risk of electrical fires, posing a serious life safety threat. This issue was observed in the coating department.

iv. Lighting tube rods and LED lights were found suspended directly from cables, inadequate supports, and several fixtures were broken due to a lack of proper support, particularly in the substation room, printing area, and sample room, posing a risk of cable strain, electrical faults, or fire hazards.F-202604-065588

Timeline	Remediation Plan
Medium Term CAP-0078054	i. Relocate power receptacles from inappropriate locations to more suitable, protected areas to prevent exposure to environmental elements. For external receptacles, utilize suitably rated IP fixtures.
Medium Term CAP-0078055	ii. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk.

Medium Term CAP-0078056	iii. Replace or repair broken and exposed power receptacles immediately to restore safe and secure connections.
Medium Term CAP-0078057	iv. Install appropriate lighting fixtures with secure mounting provisions to support bulbs and LED tube lights independently, without relying on electrical cables.
Longer Term CAP-0078058	i. Develop installation guidelines and training for personnel to ensure future power receptacles are installed in compliant and safe locations.
Longer Term CAP-0078059	ii. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.
Longer Term CAP-0078060	iii. Implement a preventive maintenance program that includes periodic checks and timely replacement of damaged electrical components to maintain safety standards.
Longer Term CAP-0078061	iv. Establish and implement a facility-wide standard for the proper installation of all lighting fixtures using durable supports, to prevent stress on electrical wiring and ensure long-term safety and reliability.





Figure 2.89 - Broken power receptacle observed in the coating department



Figure 2.90 - Inadequately supported tube rod observed in the substation room



Figure 2.91 - Power receptacle installed with wooden backing in the quilting & wadding area



Figure 2.92 - Inadequately supported lighting bulbs installed in the printing area

Figure 2.93 - Non-weather protected power receptacle installed outside building 1

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.